WEEK 7 · RESP-100 BASIC EKG INTERPRETATION

Atrial Rhythms

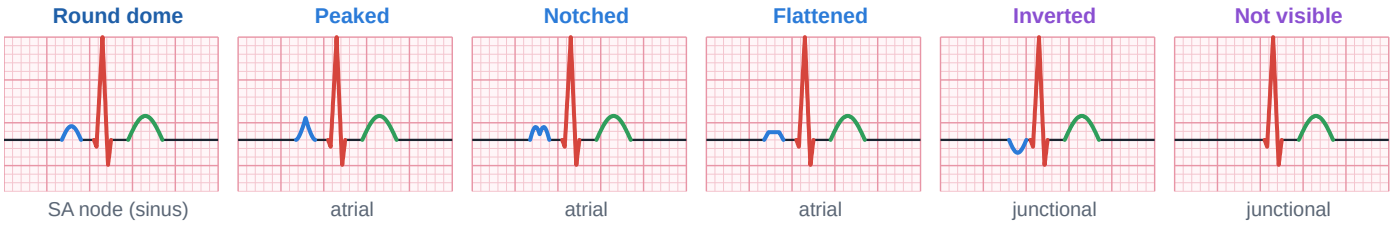
Chapter 5 · Walraven, *Basic Arrhythmias* 8th ed. · Lecture by Michael V Hyland (recording 0:00–1:08:30)

When an atrial muscle cell takes over, the QRS stays narrow. The **shape of the P wave** (or what replaces it) tells you which atrial rhythm you're looking at.

1 · Atrial basics | 2 · Wandering pacemaker | 3 · PAC | 4 · Atrial tachycardia & SVT | 5 · Atrial flutter | 6 · Atrial fibrillation | 7 · Decision flow + cheat sheet | Quiz

1 Atrial rhythms: the big picture

Big idea: any atrial muscle cell (right or left atrium) can fire. It can fire **one early beat** (a PAC) or a **whole rhythm**. The impulse starts above the ventricles, so it travels down the normal His–Purkinje pathway and the **QRS stays narrow (≤ 0.12 s)**.



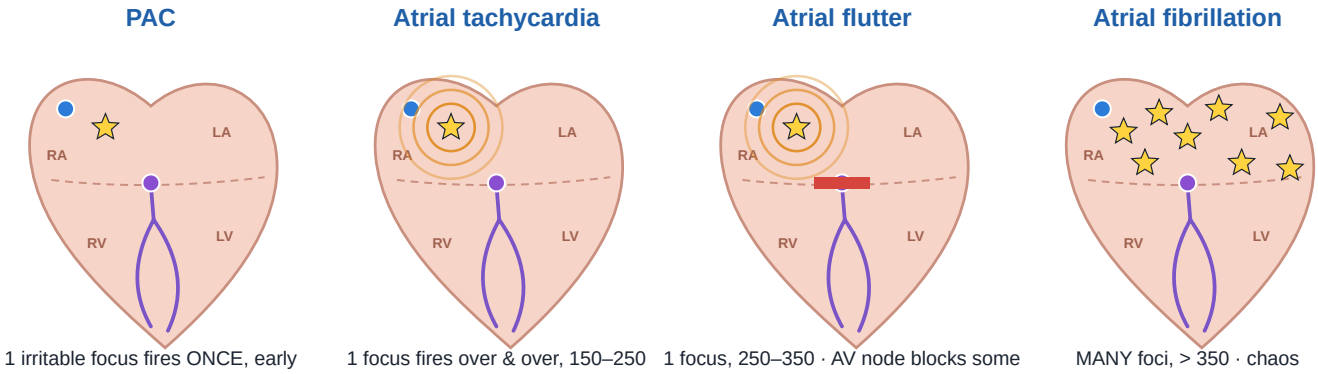
The P wave tells you where the beat started. Sinus = round dome. Atrial = peaked, notched or flattened. Junctional = inverted or not visible (Week 8).

MEMORY TRICK

Sinus = **S**mooth dome.
Atrial P waves have an "**Attitude**": **P**eaked, **N**otched or **F**lat ("PNF").
Junctional = **J**umped below the line (inverted) or gone.

IRRITABILITY → EARLY BEATS

An **irritable** cell fires **early** (premature). If it fires repeatedly and faster than the SA node, **the fastest rate wins** and it takes over.
Causes: coronary artery disease (low O_2), too much caffeine, stimulants such as cocaine.



The mechanisms at a glance. Yellow star = irritable focus, blue dot = SA node, purple = AV node + bundle branches. The red bar on the flutter heart is the AV node blocking some impulses.

Measuring reminders (from earlier weeks; you'll use them all here)

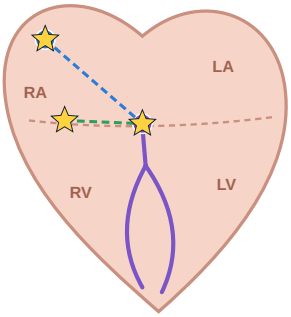
Item	The instructor's rule
Regular?	R-R gaps within **1½ small boxes** of each other = regular. More than that = irregular.
Rate, regular	**1500 ÷ small boxes** between R waves.
Rate, irregular	Count the R waves on the 6-second strip **× 10**. A half beat at each end counts as one beat.
Answers	Write **time, not boxes** (1 small box = 0.04 s). Heart rate is accepted within about **7–8 bpm** of his number.

WHAT THE INSTRUCTOR STRESSED: NEXT TEST (0:16:44)

15 strips, no multiple choice, laid out like the homework. You'll get at least **one of each**: NSR, sinus brady, sinus tach, sinus arrhythmia, a rhythm with a **PAC**, **atrial fib**, **atrial flutter**, **atrial tach**, junctional escape, PJC, accelerated junctional and junctional tach. That's 12, plus 3 of his choosing. **Wandering pacemaker will NOT be on the test.**

2 Wandering pacemaker

Big idea: the pacemaker **wanders**: SA node → an atrial cell → the AV junction and back. Each site makes a different P wave and sits a different distance from the AV node, so **the P shape AND the PRI change** from beat to beat.



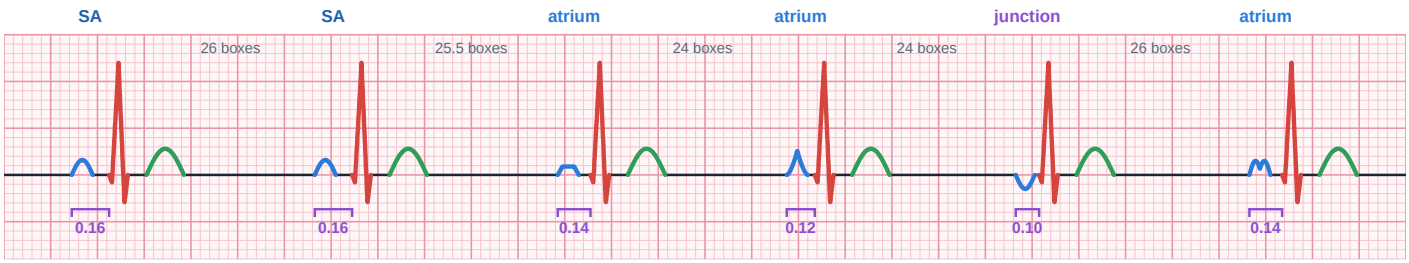
The pacemaker wanders between 3 sites

★ SA node	PRI ≈ 0.16 s
★ atrial cell	PRI ≈ 0.12 s
★ AV junction	PRI ≈ 0.08 s

Farther from AV node → longer PRI
Closer to AV node → shorter PRI

P shape AND PRI change beat to beat; QRS stays narrow.

Distance to the AV node sets the PRI (the instructor's example numbers).

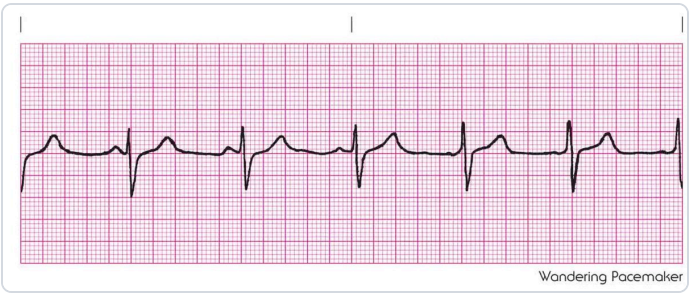


Drawn example: the label above each beat is where it started, the purple bracket under it is its PRI. R-R varies by up to 2 boxes, so it's slightly irregular.

Rules: wandering pacemaker	
Regularity	Slightly irregular
Rate	Usually normal, 60–100
P wave ★	Shape changes from beat to beat
PRI ★	< 0.20 s and **varies**. Chart it as a **range**
QRS	Narrow, ≤ 0.12 s

INSTRUCTOR (0:01:00, 0:08:30)

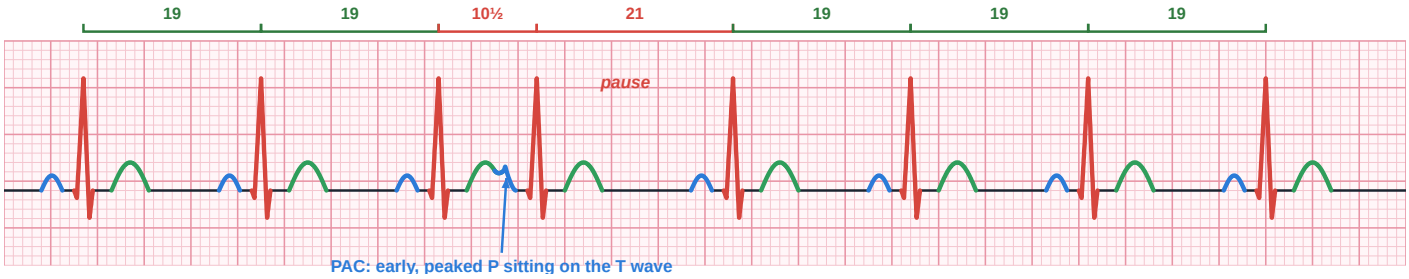
Very rare; he says he's never seen it in 25 years of ICU work. It will show up on **one homework strip** but **not on the test or final**. Put an asterisk ★ next to P shape and PRI: those are the two features that identify it.



Lecture strip (0:09:40). His worked answer:
R-R 26, 25½, 24, 24, 24½ → **irregular** (2-box spread)
6 R's × 10 → **60 bpm** · P: **shape changes**
PRIs 0.16, 0.16, 0.14, 0.10, 0.10, 0.14 → chart **0.10–0.16**
QRS 3 boxes = **0.12** → **Wandering pacemaker**

3 Premature atrial complex (PAC)

Big idea: one irritable atrial focus fires **once, early**. That gives one ectopic beat with a **different-looking P wave** and a narrow QRS. It's a **single beat, not a rhythm**, so you name the underlying rhythm too, e.g. "NSR with 1 PAC."



NSR at 78 bpm with one PAC (drawn to match the lecture strip). Green R-R gaps match, so the underlying rhythm is regular. The red gaps show the early beat (short gap) and then the pause (long gap) before the normal spacing returns.



Lecture strip (0:25:43); the arrow marks the PAC.

His worked answer

R-R: 19, 10½, 21, 19½, 19, 19… → **regular w/ 1 ectopic**
Rate: 1500 ÷ 19 = **78**
P: **upright & uniform** (the 3rd one is peaked, buried on the previous T)
PRI: 3½ boxes = **0.14** · QRS: 2½ boxes = **0.10**
NSR with 1 PAC

Rules: PAC	
Regularity	Depends on the underlying rhythm; the PAC interrupts it
Rate	Depends on the underlying rhythm
P wave	PAC's P differs from sinus P: peaked, notched or flattened. May be lost in the preceding T wave
PRI	0.12–0.20 s (can exceed 0.20)
QRS	Narrow, ≤ 0.12 s

HIS REGULARITY RULE (0:28:49)

If you can find **two R-R intervals that match** (within 1½ boxes), the **underlying rhythm is regular**, however many early beats there are. Then work out what the odd beats are.

HOW TO CHART IT

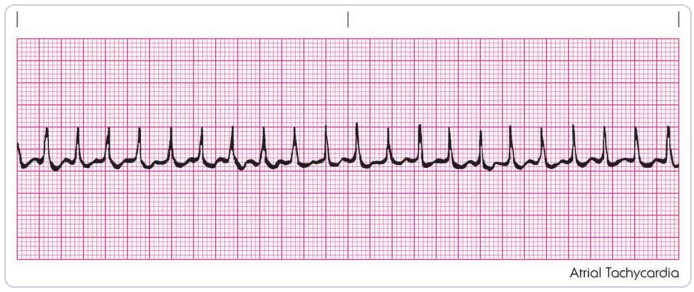
Regularity: "**regular w/ 1 ectopic**". P waves: describe the underlying rhythm ("upright & uniform"); you don't have to describe the odd P.

WHERE DID THE EARLY BEAT COME FROM? (0:36:18, 1:17:54)

Early beat + narrow QRS + **upright P that looks different** → it can only be **atrial** (a PAC). Junctional beats have no P or an inverted P. **Hidden P?** An early P can sit on the previous T, so look for a T that is **pointier or taller** than the others.

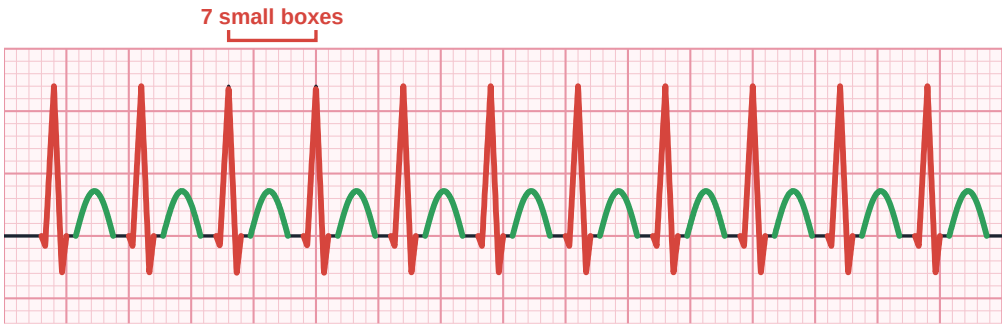
4 Atrial tachycardia & SVT

Big idea: the same kind of single irritable atrial focus as a PAC, but now it fires **over and over at 150–250 bpm** and takes over from the SA node. The AV node passes **every** impulse to the ventricles.



Lecture strip (0:41:22). R-R 7 boxes → 214 bpm, regular, P not visible, PRI can't measure, QRS 2 boxes = 0.08.

Rules: atrial tachycardia	
Regularity	Regular
Rate	150–250 bpm
P wave	Atrial P (peaked / notched / flat), but **usually lost in the T wave**
PRI	0.12–0.20 s, **only if you can see the P**
QRS	Narrow, ≤ 0.12 s



1500 ÷ 7 = 214 bpm · regular · narrow QRS · no visible P (lost in the T wave)

Drawn example of atrial tachycardia. No room for a P wave: each one is buried in the T wave before it.

Fast rates: every box counts

Small boxes R-R	1500 ÷ boxes
6	250 bpm
7	**214 bpm**
8	**187 bpm**
10	150 bpm

INSTRUCTOR (0:39:29)

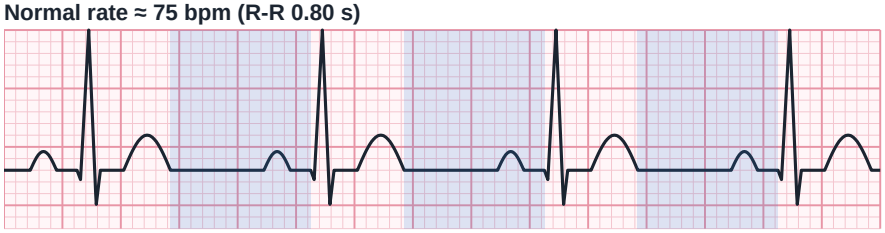
Miscount 7 boxes as 8 and you're off by 27 bpm, well outside his ~7-bpm grace. The faster the rate, the smaller your margin of error.

SVT: WHAT HE WANTS YOU TO WRITE (0:46:01)

A **narrow-complex tachycardia** with a narrow QRS comes from **above** the ventricles ("supra"). If the P waves can't be seen, you can't tell sinus vs atrial vs junctional apart, so call it **SVT (supraventricular tachycardia)**.

For this strip "atrial tach" isn't marked wrong (it's one of only two rhythms that gets above 200 in adults), but **most of the time he wants "SVT."**

Normal rate ≈ 75 bpm (R-R 0.80 s)



Atrial tachycardia ≈ 214 bpm (R-R 0.28 s)



= time for the ventricles to fill

Faster rate → less filling time → smaller stroke volume

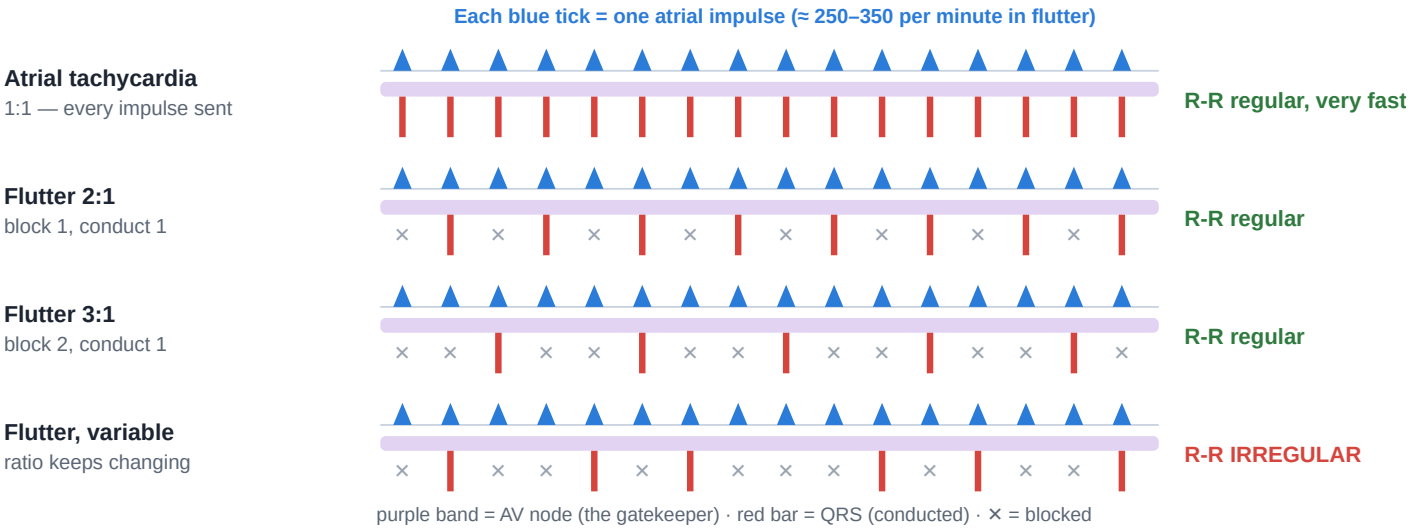
Cardiac output = HR × SV

Huge HR × tiny SV = LOW output → hypotensive, dizzy, short of breath, pale, cool & clammy, altered mental status

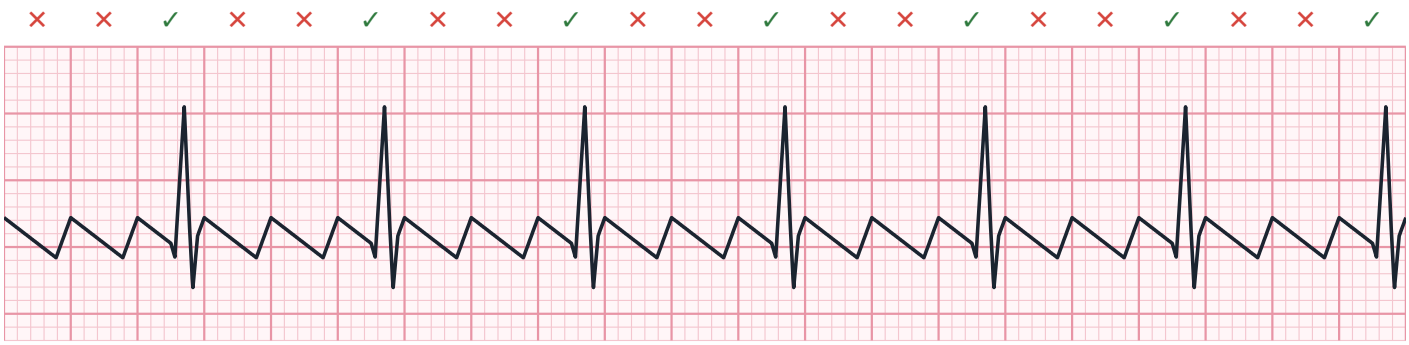
Why atrial tach is life-threatening (0:41:46): the faster the rate, the less time the ventricles have to fill (blue).

5 Atrial flutter

Big idea: one irritable atrial focus fires very fast (**250–350/min**). This time the **AV node does its job as gatekeeper**: it blocks some impulses so the ventricles aren't overwhelmed. The atrial waves look like the **teeth of a saw**.

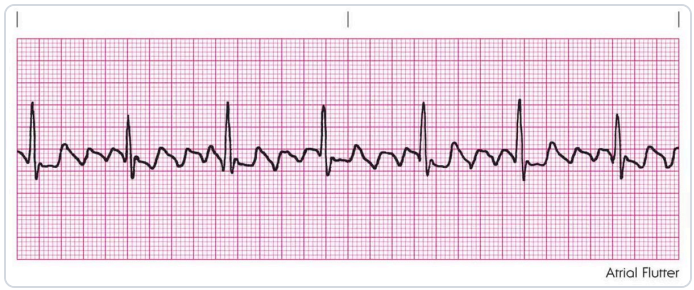


Atrial tach vs flutter is the same irritable focus; the difference is the AV node. A steady block ratio gives a regular R-R. A changing ratio gives an irregular R-R.



Sawtooth "F" waves · blocked, blocked, conducted = 3:1 · QRS narrow · PRI can't be measured

Drawn example: any P (F) wave with no QRS after it was blocked. The pattern blocked, blocked, conducted repeats, so it's 3:1 and regular.



Lecture strip (0:56:06), his worked answer: 3:1 ratio · R-R ≈ 22 → regular · 1500 ÷ 22 = **68** · P: **sawtooth** · PRI: can't measure · QRS 2½ boxes = **0.10** → **Atrial flutter**

Rules: atrial flutter	
Regularity	Ventricular (R-R) usually regular; **irregular if the block varies**
Rate	Atrial 250–350; ventricular varies (often normal)
P wave ★	**Sawtooth** pattern (F waves)
PRI	Unable to determine
QRS	Narrow, ≤ 0.12 s

INSTRUCTOR (0:55:03)

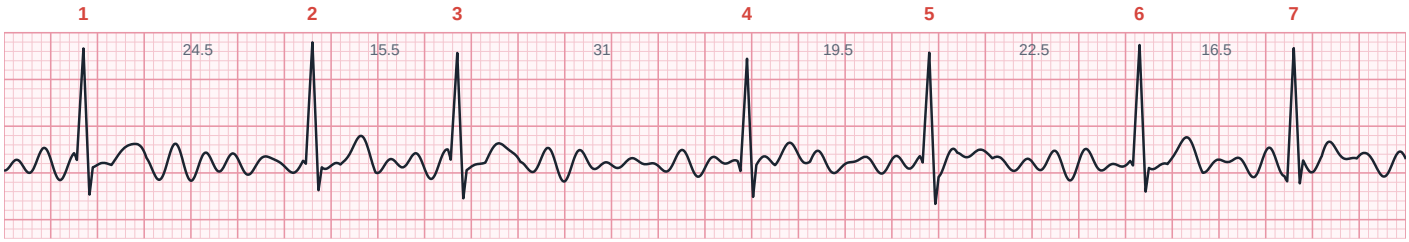
"Put an asterisk next to this P wave." If you see a **sawtooth**, it can **only** be atrial flutter. Write "sawtooth" for the P waves.

REGULARITY: VENTRICLES ONLY

He only asks for **ventricular** (R-R) regularity; you don't assess atrial regularity. Count the teeth per QRS to get the ratio (2:1, 3:1, 4:1…).

6 Atrial fibrillation (A-fib)

Big idea: the atria are so irritable that **many foci** fire at once (> 350/min). The atria can't contract; they just **quiver**. That gives no P waves (only a wavy **f-wave** baseline) and a **grossly irregular** R-R, because the AV node lets impulses through at random.



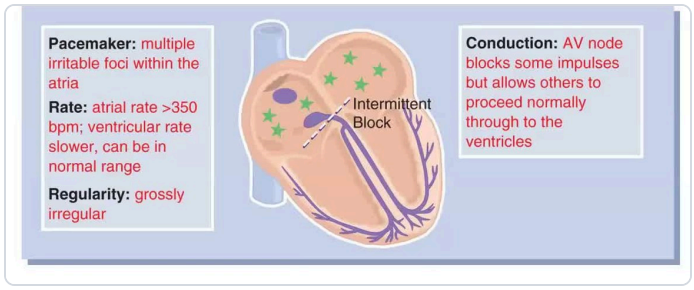
No two R-R gaps alike (grossly irregular) · wavy f-waves, no P · 7 R waves × 10 = 70 bpm

Drawn example. Every R-R gap is different (numbers = small boxes), and the baseline wiggles like artifact. Those wiggles are the fibrillatory waves.



Lecture strip (1:06:09), his worked answer: R-R 25, 31, 19½, 29½ → **irregular** · count R's × 10 = **50** · P: **f-waves** or "no P waves" · PRI: can't measure · QRS 1½ boxes = **0.06** → **A-fib**

Rules: atrial fibrillation	
Regularity	**Grossly irregular** ("irregular" is accepted)
Rate	Atrial > 350 (don't count it); ventricular varies greatly
P wave ★	None. Wavy **f (fibrillatory) waves**
PRI	Unable to measure
QRS	Narrow, ≤ 0.12 s



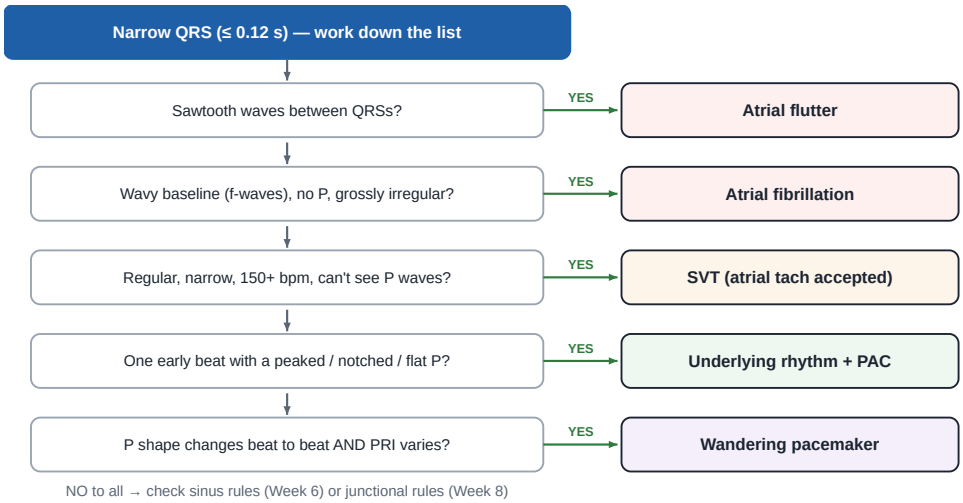
Many foci (green stars). The AV node blocks most of them (slide, 0:59:46).

CLINICAL: WHY A-FIB MATTERS (1:01:22)

No atrial contraction → less ventricular filling → smaller stroke volume.
Stagnant blood clots in the atria (like still water freezing) → a clot can travel to the brain → **stroke**. Chronic A-fib patients take **blood thinners** (e.g. Eliquis, Pradaxa).
Rate control: keep the ventricular rate **< 100**. At 140–150 filling time drops even more → hypotension.

7 Putting it together

Big idea: gather the five pieces (regularity, rate, P, PRI, QRS), then let the **P wave** (or what's there instead of it) name the rhythm.



A quick decision path for narrow-complex atrial rhythms.

Cheat sheet: Chapter 5 atrial rhythms

Rhythm	Regularity	Rate	P wave	PRI	QRS	Key clue
Wandering pacemaker	Slightly irregular	60–100	Shape changes	Varies (range)	≤ 0.12	P shape + PRI change. Not on test
PAC	Underlying + 1 ectopic	Underlying	Early P differs (peaked / notched / flat), may hide in T	0.12–0.20	≤ 0.12	Early beat, upright odd P, then pause
Atrial tachycardia	Regular	150–250	Usually hidden in T	If P seen	≤ 0.12	Can't see P → chart **SVT**
Atrial flutter	Usually regular (irregular if block varies)	Atrial 250–350; vent. varies	**Sawtooth**	Can't measure	≤ 0.12	Sawtooth = only flutter
Atrial fibrillation	**Grossly irregular**	Atrial > 350; vent. varies	f-waves / none	Can't measure	≤ 0.12	Wavy baseline + chaos

Regular, fast, no P?

≥ 150 bpm, narrow → **SVT**.

Irregular + wavy line?

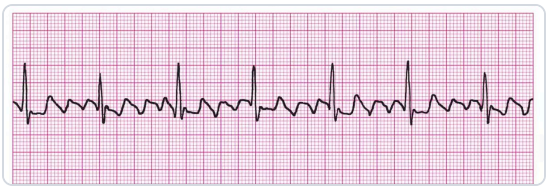
→ **A-fib**. Rate = R's × 10.

Teeth on a saw?

→ **Flutter**. Count teeth per QRS for the ratio.

? Self-check quiz

1. Name the three shapes an **atrial** P wave can have, and the shape of a **sinus** P wave.
2. In wandering pacemaker, which two findings get the asterisk, and how do you chart the PRI?
3. Why does a beat that starts in the atria still have a narrow QRS?
4. R-R intervals on a strip measure 20, 20, 11, 24, 20, 20 small boxes. Is the underlying rhythm regular? What's the rate?
5. An early beat has a narrow QRS and an upright, peaked P wave. Where did it come from? How would you name the strip if the underlying rhythm is NSR?
6. R waves are 7 small boxes apart. What's the rate? What would it be if you miscounted 8?
7. Give the instructor's preferred interpretation for a regular, narrow-QRS rhythm at 180 bpm with no visible P waves, and explain why.
8. Explain, using cardiac output = HR × SV, why atrial tachycardia can be life-threatening.
9. What's the only difference between atrial tachycardia and atrial flutter?
10. A flutter strip shows four saw-teeth for every QRS, always. What's the ratio, and is the R-R regular?
11. Give two reasons A-fib patients get medications (what kind, and why).
12. Identify strips A–D below.



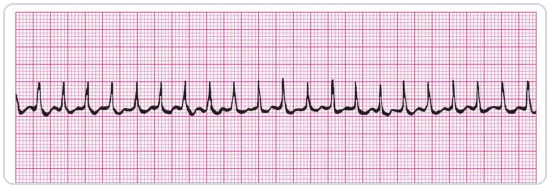
A



B



C



D

Answers. 1. Peaked, notched or flattened; sinus = round and dome-shaped. 2. P-wave shape (changes) and PRI (varies); chart the PRI as a range from shortest to longest, e.g. 0.10–0.16. 3. It still uses the normal fast His–Purkinje pathway, so ventricular depolarization is quick. 4. Yes: there are matching R-R pairs (20, 20), so the underlying rhythm is "regular w/ 1 ectopic." Rate 1500 ÷ 20 = 75. 5. The atria: an upright P that looks different can only be atrial. Name it "NSR with 1 PAC." 6. 1500 ÷ 7 = 214; 1500 ÷ 8 = 187 (a 27-bpm error). 7. SVT: narrow-complex tachycardia with no visible P, so the origin can't be told. 8. Too fast to fill → SV falls so far that CO drops despite the high HR → hypotension, dizziness, altered mental status. 9. In flutter the AV node blocks some impulses; in atrial tach it sends every impulse down. 10. 4:1, regular (the ratio stays constant). 11. Blood thinners (Eliquis, Pradaxa): stagnant atrial blood clots → stroke. Rate control: keep ventricular rate < 100. 12. A = atrial flutter (sawtooth, 3:1). B = atrial fibrillation (grossly irregular, f-waves). C = NSR with 1 PAC (the arrowed beat). D = atrial tachycardia at ≈ 214, charted as SVT.